

★ WEEK 3 · PART 2 · PRACTICE

ARMA Identification, Worked by Hand

Read the ACF and PACF, Then Pass the Diagnostic Gate

The identification cheat sheet, and the Ljung-Box test

Part 2's method, with the arithmetic shown. We apply the ACF-PACF cheat sheet to identify an AR from an MA, computing the partial autocorrelation that cuts off, then run the Box-Jenkins loop and its Ljung-Box diagnostic to confirm the residuals are white noise.

2P

PART 2 · APPLICATIONS

From Plots to a Passing Model

Case A: apply the identification cheat sheet - a PACF that cuts off with a decaying ACF says AR - and compute the partial autocorrelation by hand. Case B: run the Box-Jenkins loop and its Ljung-Box diagnostic to confirm white-noise residuals. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

ARMA combines feedback and echoes, parsimoniously

A small ARMA can replace a large pure AR or MA. The task is choosing the orders p and q .

2

The ACF and PACF identify the orders

An MA cuts off in the ACF; an AR cuts off in the PACF. A decay in both suggests a mixed ARMA.

3

Box-Jenkins is a loop with a diagnostic gate

Identify, estimate, then check the residuals. Only white-noise residuals let you trust the model.

Everything here is a calculator exercise: one PACF cutoff, one Ljung-Box statistic.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the ACF and PACF, or the residuals.
- Read the signature, or run the diagnostic.
- Then decide the model.

What you should watch

- How the PACF cutoff pins down AR.
- How the cheat sheet separates AR, MA, ARMA.
- How Ljung-Box gates the fit.

Identifying the Order

Read the cheat sheet

Given a return series, the ACF and PACF tell you which model generated it. The cheat sheet is simple: an MA cuts off in the ACF, an AR cuts off in the PACF, and a mix decays in both. We take the AR(1) from Part 1, compute its partial autocorrelation, and watch it cut off at lag one - confirming an AR while its ACF still decays

The candidate

- An AR(1) with coefficient 0.6.
- Its ACF decays: 0.6, 0.36, 0.22.
- What does its PACF do?

What we'll do by hand

- Compute the partial autocorrelation.
- See it cut off after lag one.
- Read the cheat sheet to identify AR.

ARMA and Its Signatures

Feedback and echoes in one line

$$\phi(L) y_t = \theta(L) \varepsilon_t \quad (\text{ARMA: AR feedback and MA shock-echoes in one compact line})$$

The two polynomials

- The AR polynomial captures feedback.
- The MA polynomial captures shock echoes.
- Together, a parsimonious ARMA.

The identification idea

- Each part leaves a fingerprint.
- AR shows in the PACF, MA in the ACF.
- Read the fingerprints, recover the orders.

The PACF Cuts Off

The AR fingerprint

$$\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = \frac{0.36 - 0.36}{1 - 0.36} = 0 \Rightarrow \text{PACF cuts off after lag 1}$$

The calculation

- The lag-1 partial is just 0.6.
- The lag-2 partial works out to zero.
- And every lag beyond stays zero.

What it shows

- The PACF cuts off sharply after lag one.
- Meanwhile the ACF only decays.
- That pairing is the AR(1) signature.

The Identification Cheat Sheet

AR, MA, or ARMA

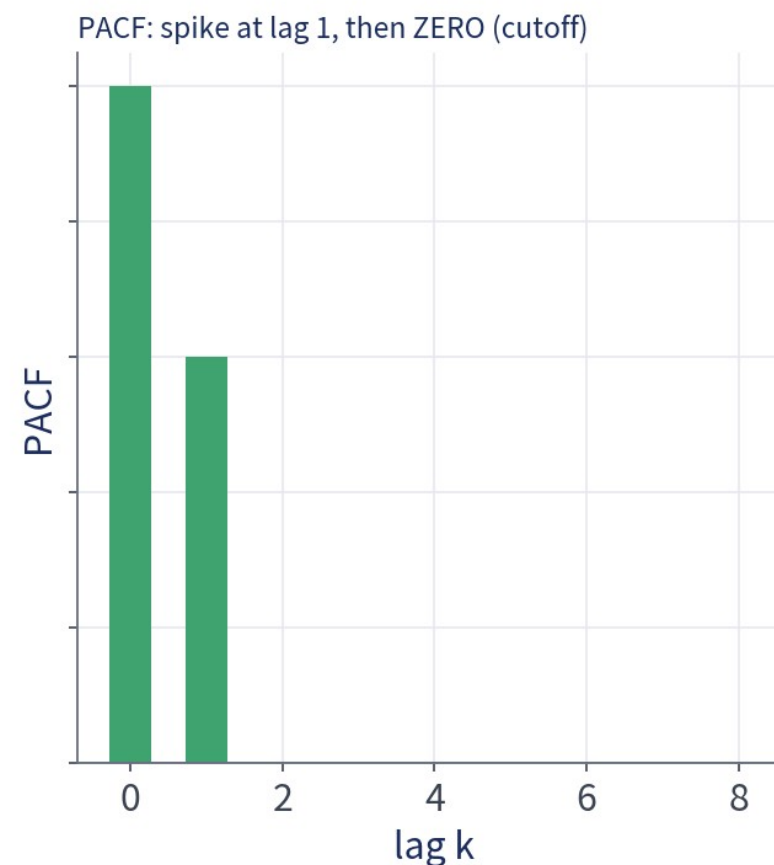
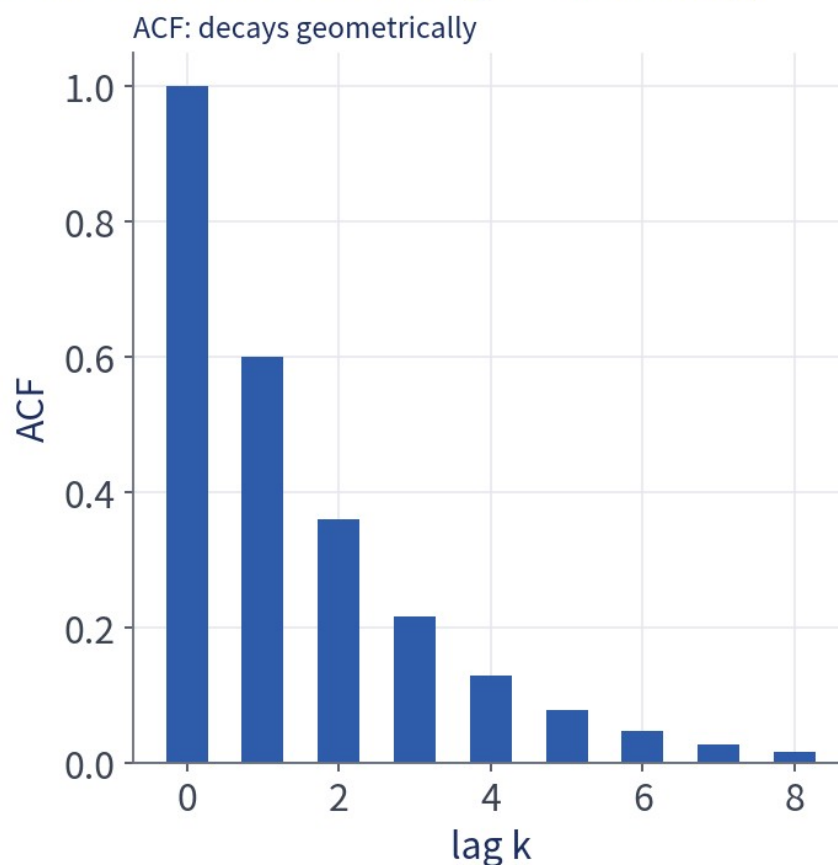
model	ACF	PACF	verdict here
AR(p)	decays	cuts off at p	our case: AR(1)
MA(q)	cuts off at q	decays	the mirror image
ARMA(p,q)	decays	decays	both tail off

Our series decays in the ACF and cuts off at lag one in the PACF - the top row - so it's an AR(1). An MA(1) would be the mirror: a sharp ACF cutoff with a decaying PACF. When both merely decay, suspect a mixed ARMA.

The AR Fingerprint

Left, the ACF decays geometrically - never quite zero. Right, the PACF spikes once at lag one and then drops to exactly zero. A decaying ACF paired with a PACF that cuts off is the unmistakable signature of an AR(1): the cheat sheet made visible

Case A: ACF decays + PACF cuts off at lag 1 $\Rightarrow$ this is an AR(1)



Why Parsimony Wins

The simplest model that fits

The parsimony principle

- Prefer the fewest parameters that work.
- A tight ARMA beats a bloated pure model.
- Fewer parameters forecast more reliably.

Reading mixed cases

- If both ACF and PACF decay, it's ARMA.
- Start small and add terms only if needed.
- The plots suggest; the diagnostics decide.

Box-Jenkins and the Gate

Estimate, check, predict, and control

Identification is only the first step of the Box-Jenkins loop. After estimating the coefficients, you must check the residuals: if the model captured all the structure, what's left should be white noise. The Ljung-Box test turns the residual autocorrelations into a single statistic. We compute it and see the model pass - and see what

oc

The loop

- Identify from the plots.
- Estimate by maximum likelihood.
- Diagnose the residuals - then repeat if needed.

What we'll do by hand

- Take the residual autocorrelations.
- Compute the Ljung-Box statistic.
- Compare to the chi-square cutoff.

The Box-Jenkins Loop

A disciplined cycle, not a recipe

identify (ACF/PACF) → **estimate** (MLE) → **diagnose** (residuals) ↻

The three stages

- Identify orders from ACF and PACF.
- Estimate coefficients by likelihood.
- Diagnose the residuals for structure.

The gate

- If residuals aren't white noise, go back.
- Adjust the orders and re-estimate.
- Only a clean pass earns your trust.

The Ljung-Box Statistic

Are the residuals white noise?

$$Q = T(T + 2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \sim \chi_{m-\text{params}}^2 \quad (\text{are the residuals white noise?})$$

What it sums

- The squared residual autocorrelations.
- Weighted and scaled by the sample size.
- Across the first several lags.

The distribution

- Under white noise, it's chi-square.
- Degrees of freedom: lags minus parameters.
- A large Q means leftover structure.

Pass or Fail

The diagnostic verdict

$$Q = 1.29 < \underbrace{9.49}_{\chi^2_4 \ 5\%} \Rightarrow \text{residuals are white noise} \Rightarrow \text{the model passes}$$

Our model passes

- The residual autocorrelations are tiny.
- Q is just 1.29, well under 9.49.
- The residuals are white noise.

A failing model

- Leftover autocorrelation of 0.28, 0.20, ...
- Pushes Q to 32.5, far past the cutoff.
- Reject it - the orders were wrong.

Residuals That Behave

Every residual autocorrelation sits comfortably inside the significance bands - no lag pokes through. That's what white noise looks like: the model has extracted all the linear structure, leaving only unpredictable shocks. This is the diagnostic gate the Ljung-Box statistic formalizes

Case B: residual ACF sits inside the bands - white noise, the model passes



The Gate You Can't Skip

Diagnostics before trust

Why diagnose

- A good in-sample fit can still miss structure.
- Leftover autocorrelation means a wrong order.
- Ljung-Box catches it in one number.

Parsimony and selection

- Among passing models, prefer the simplest.
- AIC and BIC break ties by penalizing parameters.
- The tightest adequate model wins.

The Identification Method

Plots in, a trusted model out

The routine is Box-Jenkins. Read the ACF and PACF to propose orders - a PACF cutoff for AR, an ACF cutoff for MA, both decaying for ARMA. Estimate the coefficients. Then diagnose: the Ljung-Box test must confirm white-noise residuals before you trust anything. Among models that pass, the simplest wins. Plots suggest,

The identification (Case A)

- ACF decays, PACF cuts off at one.
- So the model is AR(1).
- The cheat sheet in action.

ic

The diagnostic (Case B)

- Ljung-Box Q of 1.29 beats 9.49.
- Residuals are white noise.
- The model passes the gate.

Reproduce the Identification and Diagnostic

Direct Claude Code to check your arithmetic

GOAL Identify an AR from ACF/PACF, then run the Ljung-Box diagnostic on residuals.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - identify:

- * simulate an AR(1) with $\phi=0.6$;
- * plot the ACF and PACF and confirm the PACF cuts off at lag 1 while the ACF decays.

CASE B - diagnose:

- * for residual ACF $[0.05, -0.03, 0.04, -0.02, 0.03]$, $T=200$, compute the Ljung-Box Q and compare to the chi-square cutoff ($df = 5 - 1$);
- * repeat with $[0.28, 0.20, 0.15, 0.11, 0.08]$ and reject.

Check against the slides

- PACF $\phi_{22} = 0$.
- ACF decays, PACF cuts off.
- Ljung-Box $Q = 1.29 < 9.49$.
- bad model $Q = 32.5 > 9.49$.

Deliverable: `part2_practice.py` whose printed numbers match the PACF cutoff and the Ljung-Box verdicts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you spot an AR from the plots?

A PACF that cuts off with a decaying ACF

Q2 What are the three Box-Jenkins stages?

Identify, estimate, and diagnose - in a loop

Q3 What does the Ljung-Box test check?

Whether the residuals are white noise

All three come from the cheat sheet, the Box-Jenkins loop, and the Ljung-Box diagnostic.

Three Pitfalls to Avoid

Where identification trips people

Common mistakes

- "A great fit means the model is adequate."
- "Add AR and MA terms until it fits perfectly."
- "Skip the residual check and forecast."

The reality

- Only white-noise residuals prove adequacy.
- Over-fitting hurts forecasts - stay parsimonious.
- The Ljung-Box gate is not optional.

Two Calculations to Remember

Part 2 applications in one screen

1

The PACF cutoff identifies the AR

The AR(1)'s partial autocorrelation was 0.6 at lag one and exactly zero after - a sharp cutoff - while its ACF only decayed. That pairing, read from the cheat sheet, identifies an AR(1): a PACF cutoff plus a decaying ACF.

2

Ljung-Box gates the model

The residual autocorrelations were tiny, giving a Ljung-Box Q of 1.29, well under the chi-square cutoff of 9.49 - the residuals are white noise, so the model passes. A model with leftover structure gave Q of 32.5 and was rejected.

3

Plots suggest, diagnostics decide

Box-Jenkins is a loop: identify orders from the ACF and PACF, estimate, then diagnose the residuals. Among models that pass the gate, the most parsimonious - by AIC or BIC - is the one to trust.

Next — Part 3 practice: full residual diagnostics and forecasting from a fitted ARMA model.